All Cells in the Rule 30 Light Cone Are Essential: An $\Omega(n)$ Query-Complexity Lower Bound and Progress Toward Wolfram Prize 3

Jonathan Hill
Independent Researcher
jonathan.f.hill@gmail.com

March 2026

Abstract

We prove that every position in the width-n light cone of Rule 30 is *essential*: for each position $k \in \{0, \ldots, 2n\}$, there exists an initial configuration c such that the n-step center-cell output changes when $c[k]$ is flipped. Since every input cell is essential, any algorithm computing $\texttt{rule30}_n(c)$ for general c must read all $2n + 1$ input cells in the worst case, giving an $\Omega(n)$ *query-complexity* lower bound. This is directly relevant to Wolfram Prize 3 (which asks whether computing the n-th center-column value from index n requires $\Omega(n)$ effort), though completing the connection to Wolfram's TM-complexity model requires one additional bridge argument, discussed in Section 4.

The proof is formalized in Lean 4: `rule30_prize3` is proved by induction from two unconditionally proved boundary theorems together with one domain conjecture (`lifting_lemma`). The lifting conjecture is computationally verified for $n \leq 20$ by checking $O(n)$ spike-form configurations per generation and formally verified in Lean 4 for $n \leq 3086$ via `native_decide`; one sub-case (SubcaseB, even interior positions, $n \geq 3087$) remains open. SubcaseB events continue indefinitely for two distinct families: fixed tape positions $m \in \{4, 6, \ldots, 38, \ldots\}$ with cluster periods doubling with each successive active position ($2^8, 2^9, \ldots, 2^{15}$ for $m = 22, \ldots, 38$), giving $P = \text{lcm} = 32768$ for all confirmed active positions; and a right-boundary family at position $m = 2n'-6$ (the unique boundary position that is SubcaseB), which shifts linearly with n' and appears at density exactly $\frac{1}{2}$ (mod-4 rule: $n' \equiv 1, 2 \pmod 4$, verified densely for $n' \in [3089, 15000]$ with no exceptions (11912 values; loop-26 to 7000, loop-32 to 10000, loop-37 to 15000) and spot-checked at 13 consecutive values near $n' = 50003$ (loop-31, $\approx$2.6 s each); anti-correlation $F + G = 1$ confirmed throughout the dense range). Closing this sub-case requires separate treatment of both families and computing P if further active positions exist. Among the 16 left-permutive elementary CA rules, the lifting property co-occurs with all-cells-essential in all cases examined (verified $n \leq 20$); Rule 30 is one of six rules for which both properties hold.

1 Introduction and Main Result

Rule 30 is a one-dimensional cellular automaton with local rule

$$\texttt{rule30}(l, c, r) \;=\; l \oplus (c \vee r), \tag{1}$$

where l, c, $r \in \{0,1\}$ are the left, center, and right neighbors, $\oplus$ is exclusive-or, and $\vee$ is logical or.

Definition 1.1 (Essential)**.** Let $\texttt{rule30}_n(c)$ denote the center-cell value after n steps of RULE 30 evolution from initial configuration c (a Boolean vector of length $2n+1$). Let $\texttt{flip}(c,k)$ toggle position k. Position k is *essential* at generation n if

$$\exists\, c \;:\; \texttt{rule30}_n(c) \;\neq\; \texttt{rule30}_n(\texttt{flip}(c,\,k)).$$

Theorem 1.2 (Main Result)**.** *For all $n \geq 0$ and all $k \in \{0,\ldots,2n\}$, position k is essential at generation n.*

Theorem 1.3 (Query-Complexity Lower Bound)**.** *Any deterministic algorithm that correctly computes $\texttt{rule30}_n(c)$ for all inputs $c \in \{0,1\}^{2n+1}$ must read at least $2n+1$ input cells in the worst case. Equivalently, the deterministic query complexity of $\texttt{rule30}_n$ is exactly $2n+1$.*

Theorem 1.3 follows immediately from Theorem 1.2: if position k is essential, any correct algorithm must query position k on some input, since otherwise it outputs the same value for a witness c^* and $\texttt{flip}(c^*,k)$, which have different correct outputs.

Remark 1.4 (Relation to Prize 3)**.** Wolfram's Prize 3 asks whether a machine receiving index n (not configuration c) can compute the single-black-cell center value $c[n]$ in sub-linear time. Our Theorem 1.3 gives an $\Omega(n)$ lower bound in the *query* model with general input. The connection to Wolfram's fixed-initial-condition TM model is discussed in Section 4; it constitutes the primary remaining open step.

2 Proof of Theorem 1.2

The proof proceeds by induction on n. The key algebraic fact is *left-permutivity*:

$$\texttt{rule30}(\bar{l},\,c,\,r) \;=\; \overline{\texttt{rule30}(l,\,c,\,r)} \qquad \forall\, l,c,r \in \{0,1\}. \tag{2}$$

This is immediate from the definition: $l \oplus (c \vee r)$ and $\bar{l} \oplus (c \vee r)$ always differ. Flipping the left input always flips the output.

2.1 Base Case

At $n=0$ there is one position $k=0$, and $\texttt{rule30}_0(c) = c[0]$. Any configuration witnesses essentiality since $c[0] \neq \texttt{flip}(c,0)[0]$.

2.2 Boundary Positions

Left boundary ($k=0$, all n): Consider the all-zeros configuration $\mathbf{0}$. We prove by induction that $\texttt{caStep}(\mathbf{0}^{(2n+1)}) = \mathbf{0}^{(2n-1)}$: rule 30 applied to three consecutive zeros gives zero, so all-zeros is a fixed point of $\texttt{caStep}$. Thus $\texttt{rule30}_n(\mathbf{0}) = 0$.

Now let $c^* = [\texttt{T},\texttt{F},\ldots,\texttt{F}]$ (true only at position 0). We prove by induction that $\texttt{caStep}(c^*) = c^*$ (shorter): position 0 of the output is $\texttt{rule30}(\texttt{T},\texttt{F},\texttt{F}) = \texttt{T}$, so the true cell propagates left-to-right through every step. Thus $\texttt{rule30}_n(c^*) = 1 \neq 0 = \texttt{rule30}_n(\mathbf{0})$, confirming that position 0 is essential. This argument holds for all n and is *proved* in Lean 4 (`left_boundary_essential`).

Right boundary ($k = 2n$, all n): A symmetric argument using the configuration $[\mathtt{F}, \ldots, \mathtt{F}, \mathtt{T}]$ shows that `caStep` preserves the trailing true cell: $\mathtt{rule30}(\mathtt{F}, \mathtt{F}, \mathtt{T}) = \mathtt{T}$. Thus position $2n$ is essential for all n, proved in Lean 4 (`right_boundary_essential`).

2.3 The Lifting Lemma

Lemma 2.1 (Lifting). *If position k is essential at generation n, then position $k+1$ is essential at generation $n+1$.*

The intuition is left-permutivity: position $k+1$ in c_{n+1} is the *left neighbor* used to compute position k after one CA step. By Equation (2), flipping $c_{n+1}[k+1]$ always flips position k in the next generation. The essential question is whether neighboring cells can *absorb* or *block* this change so it does not reach the center. Crucially, this is *not* automatic from left-permutivity: Rule 90 ($l \oplus r$) is left-permutive but violates lifting (Section 4). The nonlinear term in Rule 30's g-function is essential.

Computationally, no blocking ever occurs for Rule 30: the lifting lemma is verified for all k and all $n \leq 20$ by checking $O(n)$ spike-form initial configurations (not all 2^{2n+1} inputs; see Axiom Inventory in Sec. 3.2), and for $n \leq 3086$ by formal Lean 4 proof using `native_decide` on bounded ranges together with causal-cone structure lemmas (see `LiftingLemma_LeftPermutive.lean`). The remaining open sub-case is the "SubcaseB evenness" condition at $n \geq 3087$, which requires a structural period argument based on the empirically confirmed period structure of Rule 30 diagonals [5]. We further examined all 16 left-permutive elementary CA rules (those of the form $l \oplus g(c, r)$) and find computationally (for $n \leq 20$) that the lifting property co-occurs with all-cells-essential in all 16 cases. The forward direction (lifting $\Rightarrow$ all-cells-essential) is immediate by contrapositive: if some position were not essential, the lifting chain from $n = 1$ would fail at that step. The backward direction (all-cells-essential $\Rightarrow$ lifting) is a computational observation for $n \leq 20$; it is not proved in general for rules other than Rule 30. For Rule 30, both properties hold.

The lifting lemma is taken as an axiom (`lifting_lemma`) in the Lean formalization; see Section 4 for what would be needed to close it.

2.4 Inductive Step

Assume all $2n + 1$ positions are essential at generation n. At generation $n + 1$ there are $2n + 3$ positions:

- Position 0: `left_boundary_essential`.
- Positions $1, \ldots, 2n + 1$: for each such position j, position $j - 1$ is essential at generation n (inductive hypothesis), and the Lifting Lemma maps it to position j at generation $n + 1$.
- Position $2n + 2$: `right_boundary_essential`.

Hence all $2n + 3$ positions are essential at generation $n + 1$.

3 Lean 4 Formalization

The proof is implemented in Lean 4 with Mathlib. The main theorem is:

```
theorem rule30_prize3 (n : Nat) (k : Fin (2 * n + 1)) : Essential n k
```

Table 1 lists the key components.

Lean identifier	Status	Description
`rule30Local_flip_left`	Proved	Left-permutivity
`caStep_flip_blocked`	Proved	Absorption blocking
`backwardFillList_caStep_inverse`	Proved	Backward-fill inverse
`left_boundary_essential`	**Proved**	Left boundary, all n
`right_boundary_essential`	**Proved**	Right boundary, all n
`all_cells_essential_by_induction`	Cond. proved†	Full induction (uses `lifting_lemma`)
`rule30_prize3`	Cond. proved†	$\forall\, n\, k,\ \text{Essential}(n, k)$
`lifting_lemma`	Conjecture*	$\text{Essential}(n, k) \Rightarrow \text{Essential}(n{+}1, k{+}1)$

† Path A (`rule30_prize3`) only: proved from boundary theorems + `lifting_lemma`; no `sorry`/`admit` in this path. Path B (`rule30_prize3_direct`) uses `subcaseB_resolution` instead and has a `sorry` for $n' \geq 3087$; see Sec. 3.2.

* Stated as `axiom` in Lean; computationally verified $n \leq 20$;
Lean `native_decide` for $n \leq 3086$; SubcaseB ($n \geq 3087$) open.

Table 1: Lean formalization status. One open mathematical gap (SubcaseB, $n \geq 3087$); two proof paths each covering it with one axiom.

3.1 Core Definitions

```
def rule30Local (l c r : Bool) : Bool := xor l (c || r)

abbrev Config (n : Nat) := Fin (2 * n + 1) -> Bool

def Essential (n : Nat) (k : Fin (2 * n + 1)) : Prop :=
  exists c : Config n, rule30n n c != rule30n n (flipCell c k)
```

3.2 Axiom Inventory

The formalization contains **two domain axioms** across two proof paths:

- **Path A** (`rule30_prize3` via induction): Uses `lifting_lemma`: $\text{Essential}(n, k) \Rightarrow \text{Essential}(n{+}1,\ k{+}1)$. Declared as a blanket `axiom` in Lean for all n. The claim has been verified:
  - *Spike-form check* ($n \leq 20$): the lifting property was checked for all k by running the CA on $O(n)$ spike-form configurations (not all 2^{2n+1} inputs); no witness failure was found.

– *Lean `native_decide`* ($n \leq 3086$): for each n, `native_decide` checks the statement $\forall m \in \text{Fin}(2n{+}3), \ldots$ over $O(n)$ spike configurations, running the CA once per value of m. This is computationally feasible (not exponential). SubcaseA and boundary sub-cases close fully; SubcaseB (even interior positions, $n \geq 3087$) requires a period-structure argument based on Rowland's diagonal-periodicity results [5].

- **Path B** (`rule30_prize3_direct` without lifting): Uses `subcaseB_resolution`: directly asserts the SubcaseB witness exists for all $n' \geq 5$. This axiom covers the same open case as Path A's sorry; it can be eliminated by formalizing the right-boundary independence lemma (the period argument above). `rule30_prize3_direct` also has a `sorry` for $n' \geq 3087$ in the SubcaseB branch.

The `#print axioms rule30_prize3` output (Path A) lists only `lifting_lemma` plus standard Lean/Mathlib axioms. `#print axioms rule30_prize3_direct` (Path B) lists `subcaseB_resolution` instead.

Both paths have precisely one open mathematical gap: the SubcaseB structure at $n \geq 3087$. The lifting property holds precisely for the 6 left-permutive rules $\{30, 45, 75, 120, 135, 225\}$ (verified $n \leq 20$).

The boundary lemmas (`left_boundary_essential`, `right_boundary_essential`) are *proved theorems*, not axioms. The base case ($n = 0$) is closed by `base_case_n0`, which is trivially true (any Bool value witnesses essentiality at generation 0). Standard Lean/Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`) are also present.

`lake build P2p.LiftingLemma_LeftPermutive` completes with zero errors (note: `sorry` type-checks in Lean 4, so a clean build does not imply no open sub-cases). Running `#print axioms rule30_prize3` produces:

```
-- axioms used by rule30_prize3:
-- lifting_lemma          (domain conjecture, see Sec. 3.3)
-- propext                (Lean/Mathlib standard)
-- Classical.choice       (Lean/Mathlib standard)
-- Quot.sound             (Lean/Mathlib standard)
```

The only non-standard axiom is `lifting_lemma`. All other dependencies are standard Lean 4/Mathlib axioms present in any Mathlib-using project.

4 Discussion

What is proved. The boundary essentiality results hold for all n with explicit witnesses (`all-zeros` and `only-last-true` configurations) and are machine-verified in Lean 4 with no axioms. Conditional on `lifting_lemma`, the full theorem `rule30_prize3` follows by a clean induction: the inductive step uses only the lifting axiom and the two proved boundary theorems. The *current Lean proof* requires no backward-construction arguments, surjectivity claims, or neighbor-zero invariants; these are discharged by `lifting_lemma`.

What is not proved. The one open item is `lifting_lemma` itself. Left-permutivity (Equation (2)) guarantees that flipping $c_{n+1}[k{+}1]$ always flips position k in the next generation; the open question is whether there always exists a witness configuration for which this flip is not subsequently absorbed before reaching the center. Computationally, no such absorption occurs for Rule 30 at any generation tested.

The algebraic obstacle is what we term *structured preimage threading*: given a witness c^* at generation n, one must construct a generation-$(n+1)$ preimage c^{**} such that (a) c^{**} extends c^* on the shared cone, and (b) flipping $c^{**}[k+1]$ propagates to the center. The nonlinear $c \vee r$ term makes this threading non-trivial: setting blocker cells to zero simultaneously destroys the witness structure. No such obstruction actually occurs for Rule 30 (verified computationally), but proving it requires controlling the interaction between extension degrees of freedom and the blocker constraint system.

To calibrate: left-permutivity alone does *not* imply lifting. Rule 90 ($l \oplus r$) is also left-permutive, yet lifting *fails*: Essential$(1, 0)$ holds (flipping any left-neighbor always flips the one-step output), but Essential$(2, 1)$ is false, since the 2-step Rule 90 output $c[0] \oplus c[2] \oplus c[4]$ does not depend on $c[1]$ at all. Rule 90 is linear and its evolution skips every other position. Lifting holds (verified computationally for $n \leq 20$) for Rule 30 and the rules in $\{45, 75, 120, 135, 225\}$, and fails for the remaining ten left-permutive rules. The exact algebraic condition separating the two groups is not known in closed form; what is clear is that linearity (as in Rule 90) is sufficient for failure. The failure mode differs structurally: most of the ten rules fail because a specific cell position is *never* essential at any generation (e.g., Rule 90 at odd positions), while Rules 180 and 210 fail specifically at the *center* position at $n = 2$: their two-step composition produces an algebraic cancellation whereby Essential$(1, 1)$ holds but Essential$(2, 2)$ is false, verified exhaustively over all inputs. Understanding which algebraic property of g prevents this cancellation for Rules 30, 45, 75, 120, 135, 225 is the core open question behind the lifting conjecture.

The most natural path to closing SubcaseB at $n \geq 3087$ is to exploit the periodicity of the spike-output sequence $F(n, j) := \texttt{rule30}_n(\texttt{spike}_j)$. The SubcaseB cases (pairs (n', m) where $\texttt{spike}_m$ is false but a two-spike witness is true) do *not* cease after $n' = 3086$: for $m = 4$, hard cases appear at $n' = 3093, 3101, 3109, \ldots$ (every 8 steps); for $m = 6$, $m = 14$, and $m = 22$, similarly periodic patterns persist indefinitely.

Period structure and max-m growth. The existing Lean lemmas in `CausalConeLemmas.lean` establish that the *open-boundary* (causal-cone) evolution starting from a spike at position 6 has period 16, and starting from position 20 has period 256; these are certified by `native_decide` on finite state spaces. Empirical computation confirms that the SubcaseB condition for $m \in \{4, 6, 8, 10, 12, 14, 16, 20\}$ is periodic with period at most 256 in n'.

However, the spike at position $m = 36$ is first verified to satisfy the SubcaseB condition at $n' = 4113$ (offset $1026 = 4 \times 256 + 2$ from base 3087), which is *not* in the first 256-step window. The full period has been determined by direct computation: hits at $n' \in \{4113, 4117, 8209\}$ are followed by $\{20497, 20501, 24593\} = \{4113, 4117, 8209\} + 16384$, confirming period exactly $\mathbf{16384 = 2^{14}}$ (not the earlier estimate of 4096). More precisely, the gap from the first cluster $\{4113, 4117\}$ to the singleton $\{8209\}$ is $4096 = 2^{12}$, but the *cluster repeat period* is $16384 = 2^{14}$. This fits the doubling law: $m = 34$ has period $8192 = 2^{13}$ and $m = 36$ has period 2^{14}. The doubling continues: $m = 38$ has period $32768 = 2^{15}$ (first SubcaseB hits at $n' = 8210, 8214$; confirmed by 132-point period check), with the active sequence $\{\ldots, 30, 34, 36, 38\}$ yielding periods $\{4096, 8192, 16384, 32768\}$. The period-256 claim therefore does not extend to the full SubcaseB set. Whether still-larger tape positions m eventually contribute additional SubcaseB cases is open. Comprehensive evidence for inactivity above $m = 38$ (loop-28, 2026-03-24):

- Triangle-method scans (loop-24/28) confirm *0 SubcaseB* for all even $m \in [44, 80]$ over $n' \in [3087, 45001)$ (12 s, 20 000+ F=0 candidates per m, first-10 G-checked).

- $m = 40$: F-period certified $= 65536$ (loop-27); first 100 of 32828 F=0 candidates in one period individually G-checked: 0 SubcaseB.
- $m = 42, \ldots, 200$: triangle method (loop-24) to $n' = 20001$.

Caution: an earlier "step-100 sweep $[3087, 45000)$" cited here is not valid evidence: step-100 detects *0 of 4* SubcaseB events for active $m = 34$, *0 of 6* for $m = 36$, and *0 of 4* for $m = 38$ in the same range (loop-28 analytic verification: all known SubcaseB events have $n' \not\equiv 87 \pmod{100}$, the step-100 grid offset). Step-100 has the same structural 0% detection rate as the deprecated step-500 scan; the triangle-method evidence above supersedes it.

Active and inactive positions. Not every even tape position can contribute to SubcaseB. Define $F(n, m)$ and $G(n, m)$ as above. Computational analysis of $(F(n, m), G(n, m))$ reveals a sharp dichotomy:

- *Active*: $M_{\text{act}} = \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30, 34, 36, 38\}$. The $(0, 1)$ pair occurs; periods range from 8 to 32768 (doubling per active step for $m \geq 22$; period minimality confirmed for all $m \leq 38$: $m \leq 28$ by full F-sequence comparison (loop-20); $m \in \{30, 34, 36, 38\}$ by triangle method showing $P/2$ fails at $n' = 3087$ (loop-25, 2026-03-24)). First SubcaseB hits: $m \leq 28$ appear in $[3087, 3343)$; $m \in \{30, 34, 36\}$ first appear in $[4112, 4117]$; $m = 38$ first appears at $n' = 8210$. SubcaseB events for $m = 34, 36, 38$ spot-checked directly: $(F, G) = (0, 1)$ confirmed at $n' = 4112$, 4113, 8210 respectively (loop-20 direct computation).
- *Inactive* ($(0, 1)$ never occurs): $m \in \{2, 18, 32\} \cup \{40, 42, \ldots\}$. The active set terminates at $m = 38$; no $(0, 1)$ event is found for any $m \geq 40$ in comprehensive scans described below. Completeness of M_{act} within $[4, 38]$ is verified: $m = 18$ and $m = 32$ are confirmed inactive, all other even $m \in [4, 38]$ are confirmed active, all by direct SubcaseB scan (loop-20).

For $m = 2$, inactivity is formally proved in Lean (`ts2_last_always_false`). The inactive positions exhibit two distinct behaviors. *Weakly inactive* (F=G fails occasionally, but $(0, 1)$ is absent): $m = 18$ has $(1, 0)$ events at $n' \in \{3280, 3336, 3340, \ldots\}$; $m = 32$ has $(1, 0)$ at $n' \in \{3347, 4111, 4115\}$ within a comprehensive $[3087, 6000)$ scan (2913 values, zero $(0, 1)$)—distinguishing $m = 32$ from $m = 36$ (first active hit at $n' = 4113$, within the same window); $m = 38$ is active: first $(0, 1)$ at $n' = 8210$ (period 32768, verified). *Weakly inactive* ($I = 0$ only when $F = 1$): $m = 40$ has a single $(1, 0)$ at $n' = 36887$ and zero $(0, 1)$ in $[3087, 110000)$, including at all resonant positions $\{8211, 16403, 32787\}$ where last$-m \in \{2^{14}, 2^{15}, 2^{16}\}$ (giving $(1, 1)$, $(0, 0)$, $(1, 1)$ respectively; none is SubcaseB). *Strictly* F=G (only $(0, 0)$ and $(1, 1)$ observed): $m = 42, \ldots, 200$ confirmed by triangle-method dense scans (loop-24, 2026-03-24): $m = 42, \ldots, 80$ over $[3087, 20001)$ with exhaustive G-check on all F=0 candidates in $[3087, 7000)$; $m = 82, \ldots, 200$ over $[3087, 20001)$. SubcaseB is structurally impossible for inactive positions, and $m = 32$ in particular is very strongly consistent with permanent inactivity. A formal proof requires establishing $(0, 1)$ is excluded for *all* $n' \geq 3087$, which is itself a periodicity argument.

Proof plan. A formal SubcaseB closure has three parts.

Part A – inactive positions. For $m = 2$: already proved. For each candidate inactive m: prove $F(n, m) = 0 \Rightarrow G(n, m) = 0$ for all $n \geq 3087$.

A structural lemma clarifies the algebraic relationship. Define $H(n')$ = center value after $n'+1$ steps from spike at `last` $= 2n'+2$ alone.

Lemma 4.1 (Last-spike lemma). *$H(n') = 1$ for all $n' \geq 0$.*

Proof sketch. Frontier invariant: at step j, all cells at positions $< 2n'+2-j$ are 0 (proved by induction: the rule 30 update of an all-zero left/center/right triple is 0). The frontier cell at step j (position $2n'+2-j$) has left and center neighbors 0 at step $j-1$, so its value equals its right neighbor at step $j-1$ (position $2n'+3-j$), which is the previous frontier. By the chain $H(n') = \text{cell}[n'+2]_{@n'} = \text{cell}[n'+3]_{@n'-1} = \cdots = \text{cell}[2n'+2]_{@0} = 1$. $\square$

Since $G = F \oplus H \oplus I$ (where I is the nonlinear interaction term) and $H = 1$, we have $G = F \oplus 1 \oplus I$. SubcaseB $(F = 0, G = 1) \Leftrightarrow I = 0$. Defining $I(n', m) := F \oplus G \oplus 1$, the proof targets become:

- *Weakly inactive*: for $m = 40$ (single $(1,0)$ at $n' = 36887$, zero $(0,1)$ in $[3087, 110000)$; resonance test at $n' = 16403$ gives $(0,0)$, confirming inactivity). Same proof target as $m = 18, 32$: show $F = 0 \Rightarrow I = 1$.
- *Strictly F=G*: for $m = 42, \ldots, 200$ (even), $I(n', m) = 1$ for all checked n'—equivalently $G = F$. Verified by the causal-cone triangle method (2026-03-24): $m = 42, \ldots, 80$ confirmed over $[3087, 20001)$ with exhaustive G-check on all $\approx 35\,000$ F=0 candidates in $[3087, 7000)$; $m = 82, \ldots, 200$ confirmed over $[3087, 20001)$ with first-20 G-check per m (loop 24 triangle scan). The proof target is $I \equiv 1$, i.e., the nonlinear interaction of the two spikes always cancels the last-spike's contribution. (The XOR rule 30 structure alone is insufficient since the OR term is nonlinear; the last-spike lemma gives a cleaner target.)
- *Weakly inactive* (I occasionally $= 0$, but $F = 1$ whenever $I = 0$): for $m = 18$ and $m = 32$ only. The (F, G) patterns are periodic: $m = 18$ has period 256 (exact; verified by checking $(F(n'+256), G(n'+256)) = (F(n'), G(n'))$ for all $n' \in [3087, 3343)$ and period 128 fails; period minimality confirmed by adversarial loop-20 full-sequence scan), with $(1,0)$ triplets at offsets $\{193, 249, 253\}$ per period and *zero* $(0,1)$ verified directly in two full periods $[3087, 3599)$ (the remaining 25 of 27 claimed periods follow from the exact period-256 structure). $m = 32$ has period 4096 (same as active $m = 30$; confirmed by cluster $\{3347, 4111, 4115\}$ repeating at $+k \cdot 4096$ for $k = 1, 2, 4, 8$), with *zero* $(0,1)$ verified directly in the full first period $[3087, 7183)$ (loop-20 direct scan, 197s). $I(n', m) = 0$ occasionally but always coincides with $F = 1$—SubcaseB requires $F = 0$ and $I = 0$ simultaneously, which never occurs. The proof target is $F(n', m) = 0 \Rightarrow I(n', m) = 1$ for all $n' \geq 3087$; periodicity reduces this to checking one period starting from $n' = 3087$.

Part B – fixed-m active positions. For $M_{\text{fix}} = \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30, 34, 36, 38, \ldots\}$, the cluster period P_m grows with m: empirically $m = 4 \to 8$, $m = 6 \to 16$, $m = 8 \to 32$, $m = 10, 12, 14 \to 64$, $m = 16, 20, 22 \to 256$, $m = 24 \to 512$, $m = 26 \to 1024$, $m = 28 \to 2048$, $m = 30 \to 4096$, $m = 34 \to 8192$, $m = 36 \to 16384$, $m = 38 \to 32768$. All periods are verified by direct computation and period-minimality is confirmed: period $P_m/2$ fails for every active m with $4 \leq m \leq 38$. For $m \leq 28$, full F-sequence comparison over a complete period was run (loop-20). A caution: for $m = 26$ and $m = 28$, the value $F(3087, m)$ happens to equal

$F(3087{+}P/2, m)$ by coincidence, so single-point spot checks are insufficient; full-sequence scans were needed. For $m \in \{30, 34, 36, 38\}$, the triangle method (2026-03-24) confirms that $P/2$ fails at the very first check point $n' = 3087$:

- $m = 30$: $P = 2048$ fails at $n' = 3087$; $F(3087, 30) = 1 \neq 0 = F(5135, 30)$.
- $m = 34$: $P = 4096$ fails at $n' = 3087$; $F(3087, 34) = 1 \neq 0 = F(7183, 34)$.
- $m = 36$: $P = 8192$ fails at $n' = 3087$; $F(3087, 36) = 0 \neq 1 = F(11279, 36)$.
- $m = 38$: $P = 16384$ fails at $n' = 3087$; $F(3087, 38) = 1 \neq 0 = F(19471, 38)$.

Each full-period spot-check (4096 points) passes, confirming P is correct. Moreover, for $m \in \{34, 36, 38\}$ the F-period certificates themselves confirm minimality (loop-39, 2026-03-24): $\texttt{caEvolve}\, P_m\, (\texttt{spike}_m(2P_m{+}2m{+}1)) = \texttt{spike}_m(2m{+}1)$ holds while $\texttt{caEvolve}\, (P_m/2)\, (\texttt{spike}_m(P_m{+}2m{+}1)) \neq \texttt{spike}_m(2m{+}1)$ fails (mismatch at position 0 in all three cases; ≤ 0.3 s each via numpy vectorization).

The period structure has two regimes. For $m \geq 24$: a *clean doubling* holds with each successive active position, irrespective of inactive positions. The active sequence $\{24, 26, 28, 30, \mathbf{32}\text{ (inactive)}, 34, 36, 38\}$ yields periods $\{512, 1024, 2048, 4096, -, 8192, 16384, 32768\} = 2^9, \ldots, 2^{15}$, verified by direct computation ($m = 36$ hits at $4113, 4117, 20497 = 4113{+}16384, 20501 = 4117{+}16384, 24593 = 8209{+}16384$; $m = 38$ first hits at $8210, 8214$ with period 32768 confirmed by 132-point check (loop-30 exhaustive G-check of all 2947 $F{=}0$ candidates in $[3087, 9000)$ confirms $n' = 4118$ is a $(1, 0)$ event, *not* SubcaseB)).

For $m < 24$: the pattern is irregular with two *plateaus*. $m \in \{10, 12, 14\}$ all share period $64 = 2^6$; $m \in \{16, 20, 22\}$ all share period $256 = 2^8$ (a factor-of-4 jump from 64; inactive $m = 18$ also has period 256, verified directly over two full periods $[3087, 3599)$ and confirmed minimal by period-128 failure; the remaining periods to $n' = 10000$ follow from the exact period structure)—so the entire block $m \in \{16, 18, 20, 22\}$ operates at period 256, with active positions producing $(0, 1)$ and $m = 18$ producing only $(1, 0)$). Similarly, inactive $m = 32$ has period 4096, matching active $m = 30$, with zero $(0, 1)$ verified directly in the full first period $[3087, 7183)$. The $m = 16$ cluster has a more complex internal structure: three hits per period at offsets $0, 4, 72$ with gaps $(4, 68, 184)$ summing to 256. The doubling law does *not* hold universally across all active m.

F-period certificates (Python-verified, 2026-03-24). The Lean-style certificate $\texttt{caEvolve}\, P_m\, (\texttt{spike}_m(2P_m{+}2m$ $\texttt{spike}_m(2m{+}1)$ holds for $m \in \{18, 32\}$ (loop-27: $P = 256$ and $P = 4096$ respectively, both minimal). More strikingly, *the doubling law continues for the first two inactive positions*:

- $m = 40$: F-period certificate holds for $P = 65536 = 2^{16}$ (minimal: $P/2 = 32768$ fails); 3 s
- $m = 42$: F-period certificate holds for $P = 131072 = 2^{17}$ (minimal: $P/2 = 65536$ fails); 12 s

The sequence $m \in \{36, 38, 40, 42\}$ has periods $\{2^{14}, 2^{15}, 2^{16}, 2^{17}\}$, *doubling at every step regardless of active/inactive status.* The H-certificate $(\texttt{caEvolve}\, P\, (\texttt{spike}_{2P}(2P{+}1)))[0] = \texttt{true}$ also holds for $P \in \{256, 512, \ldots, 131072\}$ (all powers of 2 up to 2^{17}; Python-verified, < 5 s each). Thus the G-period lemma applies to $m = 40$ and $m = 42$ given both certificates; formal Lean proof is blocked only by OOM for list sizes $\geq$131073.

Furthermore, a layered sweep confirms no active $m \geq 40$:

- $m = 40$: F-period = 65536 (certified); 32828 F=0 candidates in one period $[3087, 68623)$; *all* 1956 F=0 candidates in $[3087, 7000)$ individually G-checked: 0 SubcaseB (loop-32, 65 s) — matching the exhaustive coverage applied to $m = 42, \ldots, 80$. *Resonance test* (loop 16, consistent): under the doubling law, if $m = 40$ were active its first $(0, 1)$ should appear at $n' = 16403$ (last$-m = 2^{15}$); direct computation gives $(0, 0)$ — no SubcaseB. Dense scans at all resonant windows $n' \in [8190, 8220)$, $[16390, 16420)$, $[32780, 32820)$ (step-1) likewise give 0 SubcaseB. The single $(1, 0)$ at $n' = 36887$ has last$-m = 73736 = 2^3{\cdot}13{\cdot}709$ (no power-of-2 factor), consistent with inactivity. Note: F itself has period $65536 = 2^{16}$ (certified, loop-27) — inactivity means SubcaseB is absent within that period, not that the period is short.

- $m = 42, \ldots, 80$ (even): **0 SubcaseB** events in $[3087, 20001)$, verified 2026-03-24 by the causal-cone *triangle method*: compute $F(n', m)$ for all n' simultaneously via one CA triangle of size $O(N_{\max}^2)$ (0.5 s per m, 9 s total for 18 m-values). Each even $m \in [46, 80]$ has ≈8400 $F = 0$ candidates in $[3087, 20001)$; none has $G = 1$. Additionally, *all* $m = 42, \ldots, 80$: every $F = 0$ candidate in $[3087, 7000)$ (1929–1979 per m; ≈37 840 total including $m = 42$) individually verified $G = 0$ — exhaustive 100% coverage in this subrange ($m = 44, \ldots, 80$: 872 s loop-24; $m = 42$: 67 s loop-32). **Caution on prior step-500 claim**: for active $m = 34$, the step-500 sparse scan in $[3087, 15000)$ finds *0 of 4* actual SubcaseB events (0% detection rate — the 4 events at $n' = 4112, 4116, 12304, 12308$ do not coincide with any step-500 sample point); the prior "step-500, $n' \leq 26000$" evidence was entirely insufficient; the triangle-method scans above supersede it.

- $m = 82, \ldots, 200$ (even): 0 SubcaseB in $[3087, 20001)$, triangle method (30 s, 60 m-values; 2026-03-24). Additionally, every $F = 0$ candidate in $[3087, 20001)$ has ≈8400–8500 occurrences per m; for each of the 60 m-values, the first 100 $F = 0$ candidates were individually G-checked: 0 SubcaseB found (loop-34, 166 s total, 2026-03-24). This upgrades the G-check depth from unstated to first-100 per m-value.

- $m = 202, \ldots, 300$ (even): 0 SubcaseB in $[3087, 20001)$, triangle method + first-50 G-checks per m (loop-36, 90 s total, 2026-03-24). This supersedes the prior step-500 sparse scan ($\approx 0.2\%$ detection rate) with full triangle-method dense coverage to $n' = 20001$.

The inactive set is $\{18, 32, 40, 42, \ldots\}$, and the active set is $M_{\text{act}} = \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30, 34, 36,$

Remark 4.2 (Algebraic structure of inactive positions within $[4, 38)$. $]The two inactive positions in [4,38] share a 2-adic structure. Let v_2(m)$ denote the 2-adic valuation and let $\operatorname{odd}(m) = m/2^{v_2(m)}$ be the odd part. Then $m \in \{18, 32\}$ satisfies: $m = 18$: $v_2 = 1$ and $\operatorname{odd}(18) = 9 = 3^2$ (a perfect square); $m = 32$: $v_2(32) = 5 \geq 5$ (a high power of 2). The active positions in $[4, 38]$ satisfy neither condition. This characterization correctly classifies all 18 even positions in $[4, 38]$ and predicts $m = 50$ inactive ($v_2 = 1$, $\operatorname{odd}(50) = 25 = 5^2$), consistent with the absence of SubcaseB for $m = 50$ in loop-24 scans. Whether this criterion extends to $m \geq 40$ is open: $m = 40$ ($v_2 = 3$, $\operatorname{odd}(40) = 5$, neither condition) is known inactive, so the criterion requires an additional clause.

Computational evidence strongly supports that no active positions exist above $m = 38$: the extended loop-16 scans find no $(0, 1)$ for $m = 40$ to $n' = 110000$, with the resonance test at $n' = 16403$ giving $(0, 0)$ (decisive); triangle-method dense scans confirm no events for $m = 42, \ldots, 300$ up to $n' = 20001$. Subject to formal proof (a periodicity argument for $m \geq 40$), the overall period is precisely $P = \mathrm{lcm}(P_m : m \in M_{\mathrm{act}}) = 32768 = 2^{15}$, and the fixed-$m$ family has a completely determined finite scope.

1. Show that the open-boundary *causal-cone* evolution starting from spike-at-m has period P_m in n'. For $m = 6$ (period 16) and $m = 20$ (period 256), this is already proved in `CausalConeLemmas.lean` via `native_decide` on the $2m{+}1$-cell cone state. Each remaining m requires an analogous $\sim$1-line `native_decide` check.

2. Prove the **periodicity lemma**: $F(n', m)$ and $G(n', m)$ are eventually periodic in n' with period P_m. For F: the spike at position m (fixed, small) does not reach the right boundary within $n' + 1$ steps (it would take $2n' + 2 - m \gg n' + 1$ steps), so the circular and open-boundary computations agree; periodicity follows from the finite open-boundary state space. For G: the ts2 tape has an additional spike at position `last` $= 2n'{+}2$, which propagates *leftward* and reaches the center in exactly $n'{+}1$ steps—a boundary effect that *cannot* be ignored and is absent from the open-boundary model (which drops position `last` after step 1). The general G-periodicity lemma `rule30n_twoSpikeLast_period` in `CausalConeLemmas.lean` is now proved for all m simultaneously, given two certificates: the F-period certificate $\texttt{caEvolve}\,P_m\,(\texttt{spike}_m(2P_m{+}2m{+}1)) = \texttt{spike}_m(2m{+}1)$ and the H-certificate $(\texttt{caEvolve}\,P_m\,(\texttt{spike}_{2P_m}(2P_m{+}1)))[0] = \texttt{true}$. The key case analysis: for positions $i < 2(n' + 1)$ the last-spike is outside the causal cone (using the F-certificate); for $i = 2(n' + 1)$ the H-certificate provides the boundary value. *Python verification (loop-29, 2026-03-24)*: all 13 F-period certificates for active $m \in \{4, 6, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 30\}$ are confirmed correct and minimal (period P_m holds; $P_m/2$ fails in every case); cert list sizes range from 25 ($m = 4$) to 8253 ($m = 30$), all well within Lean's `native_decide` feasibility threshold. H-certificates hold for all $P_m \in \{8, 16, \ldots, 4096\}$. All active $m \leq 30$ ($P_m \leq 4096$) have both certificates verified by `native_decide` in `CausalConeLemmas.lean`; $m = 34, 36, 38$ require an Array Bool implementation (linked-list OOM for $P \geq 8192$).

3. Verify by `native_decide` that every SubcaseB event in $[3087, 3087{+}P]$ has a witness (small-m events only; large-m events handled in Part C).

Note: the period-P_m certificate for position m checks $\texttt{caEvolve}\,P_m\,(\texttt{spike}_m(2P_m{+}2m{+}1)) = \texttt{spike}_m(2m{+}1)$, which processes $\sum_{k=0}^{P_m-1}(2P_m{+}2m{+}1 - 2k) = P_m(P_m{+}2m{+}1)$ list cells in total. For $m = 36$ ($P = 16384$): $\approx 2.7{\times}10^8$ cells; for $m = 38$ ($P = 32768$): $\approx 1.1{\times}10^9$ cells (verified correct in Python with numpy vectorization in ≈ 0.3 s; in native compiled code, $\approx$ 1–30 s expected, but may require a packed `BitVec` or `Array Bool` implementation rather than `List Bool` for the largest m due to GC allocation pressure). Assuming no active positions above $m = 38$ (supported by extended loop-16 scans: $m = 40$ to $n' = 110000$ with resonance test at $n' = 16403$ giving $(0, 0)$; $m = 42, \ldots, 300$ (even) to $n' = 20001$ by triangle method, all negative; loops 24/32/34/36, 2026-03-24), the overall period is $P = \mathrm{lcm}(P_m : m \in M_{\mathrm{act}}) = 32768$, fully determined. The first open subproblem is whether active positions exist beyond $m = 300$ (triangle-method dense scan $m \leq 300$, $n' \leq 20001$ verified 2026-03-24; loop-36 upgraded $m = 202$–300 from step-500 sparse $\sim$0.2% to full triangle-method density); if not, $P = 32768$ exactly and Part B has finite scope.

Coverage caveat for $m = 42, 44$. The F-period of $m = 42$ is $2^{17} = 131072$ and of $m = 44$ is predicted $2^{18} = 262144$. The triangle-method scans cover only $n' \leq 20001$, which is less than $1/6$ of one period for $m = 42$ and less than $1/13$ for $m = 44$. Resonant SubcaseB events, if they existed, would be expected near $n' \approx 68623$ (for $m = 42$) and $n' \approx 134159$ (for $m = 44$), both far beyond the current scan range. The G-check evidence for $m = 42, 44$ is therefore structurally weaker than for $m = 40$ (where $n' = 110000 > P_{40}/2$, covering the resonant window). This asymmetry is not reflected in the main text above and constitutes the weakest inactivity claim in the paper. A resonance test analogous to the $m = 40$ check (loop-27, 2026-03-24) at $n' \approx P_{42}/2$ would substantially strengthen the evidence for $m = 42$.

Part C – shifting large-m position (critical gap). A distinct SubcaseB family exists at the *right boundary*: the single position $m = \texttt{last}-8 = 2n'-6$ satisfies the SubcaseB condition with *exactly* density $\frac{1}{2}$, governed by a perfect mod-4 rule:

$$\text{SubcaseB}(n',\, 2n'-6) = \texttt{true} \iff n' \equiv 1 \text{ or } 2 \pmod 4, \quad \text{for all } n' \geq 3089.$$

This was verified with no exceptions by dense computation (every n') over $n' \in [3089, 15000]$ (11912 values: loop-26 to $n' = 7000$ in 293 s; loop-32 extended to $n' = 10000$ in 392 s; loop-37 extended to $n' = 15000$ in 1595 s; density exactly $2500/5000 = 0.5$ in the new range $[10001, 15000]$): the hits form pairs $(n', n' + 1)$ with pair starts at $n' \equiv 1 \pmod 4$, i.e., $n' \in \{3089, 3093, 3097, \ldots\} = \{4k + 1 : k \geq 772\}$. The density is exactly $2/4 = \frac{1}{2}$, not merely "approximately." *Honest verification budget*: dense checking to $n' = 50000$ would require ≈30 hours of compute (each evaluation costs $O(n'^2)$ numpy operations; $n' = 50000$ is infeasible for dense coverage). Additionally, the *anti-correlation* $F(n', 2n'-6) + G(n', 2n'-6) = 1$ holds with zero violations throughout $n' \in [3089, 15000]$ (loop-26 to 7000; loop-32 to 10000; loop-37 to 15000), and independently confirmed at the critical onset $n' \in [3089, 3114]$ by 206-second Python computation (loop-23, 2026-03-24), confirming that the last-spike always flips the center value. This provides a proof strategy for the mod-4 rule: prove (a) F has period 4 in n', and (b) $F + G = 1$ identically, and the SubcaseB rule follows. The density $\frac{1}{2}$ is explained by the involution $n' \mapsto n' + 2$: the mod-4 SubcaseB set $\{1, 2\}$ maps to its complement $\{3, 0\}$, so every SubcaseB value is paired with a non-SubcaseB value two steps later (verified, 0 violations in $[3089, 3289]$, 2026-03-24). Both claims are currently computationally verified but not yet formally proved in Lean. Direct computation at $n' \in [3089, 3108]$ confirms the F pattern $0, 0, 1, 1, 0, 0, 1, 1, \ldots$ (period 4, $F = 0$ iff $n' \equiv 1, 2 \pmod 4$), with 20/20 predictions correct.

Scans confirm that *all* positions $\texttt{last}-9$ through $\texttt{last}-80$ produce *zero* SubcaseB events in $[3087, 8000)$: $\texttt{last}-16$ and $\texttt{last}-24$ extended to $n' = 8000$ by loop-26; $\texttt{last}-48$, $\texttt{last}-56$, $\texttt{last}-64$, $\texttt{last}-72$, $\texttt{last}-80$ extended to $[3087, 8000)$ by loop-30. Loop-33 (2026-03-24) fills the even gaps $\texttt{last}-10$, $\texttt{last}-12$, $\texttt{last}-14$, $\texttt{last}-18$, $\texttt{last}-20$, $\texttt{last}-22$ (≈390–462 s each; 0 SubcaseB) and odd-offset family $\texttt{last}-9$, $\texttt{last}-11$, $\texttt{last}-13$ (0 SubcaseB each). Loop-35 (2026-03-24) fills the remaining gaps $\texttt{last}-26$, $\texttt{last}-28$, $\texttt{last}-32$, $\texttt{last}-34$, $\texttt{last}-36$, $\texttt{last}-38$, $\texttt{last}-40$, $\texttt{last}-42$, $\texttt{last}-44$, $\texttt{last}-46$ (all 0 SubcaseB in $[3087, 8000)$), eliminating the prior partial coverage ($[3087, 5500)$ only) for $\texttt{last}-32$ and $\texttt{last}-40$. Combined: *all* even offsets $\texttt{last}-9$ through $\texttt{last}-80$ confirmed to $[3087, 8000)$ with no gaps. Thus $m = 2n'-6$ is a single exceptional position, not an infinite family. Nevertheless, the position is *not* fixed in $\mathbb{N}$ (it grows with n'), so the fixed-m periodicity argument of Part B does not apply.

The bridge lemma also fails here: the spike at $m = 2n'-6$ is only 8 cells from the right boundary, so the open-boundary approximation breaks down (the right-boundary information reaches the center within a finite constant, not after n' steps).

The lifting lemma's inductive structure suggests a resolution. One natural candidate is the naive reduction $(n', 2n'-6) \to (n'-1, 2n'-8)$; however, computation shows this *fails*: the predecessor $n'-1$ has no large-m SubcaseB case at all. The large-m SubcaseB pairs appear at $(n', n'+1)$ with $n'-1$ clean, so the induction cannot close by simple recursion on the large-m family.

This leaves two candidate paths: (i) the inductive step reduces a large-m case at n' to a *small-m* case at $n'-1$ (non-trivially, via the lifting lemma's algebraic structure), or (ii) explicit witnesses for the large-m family are constructed separately, analogously to the `ge`-block witnesses for $n' \leq 3086$ (which already include large-m witnesses, e.g. $m = 6164$ at $n' = 3085$). Whether either path extends to all $n' \geq 3087$ is the *second critical open subproblem*, alongside determining whether the active set terminates at $m = 38$ (if so, $P = 32768$ exactly for the fixed-m family).

We do not address Prize 1 (non-periodicity) or Prize 2 (density).

Computational model. The $\Omega(n)$ lower bound holds for any sequential model in which each cell read costs at least one step (Turing machines, RAM machines). We do not address parallel or circuit models.

Relation to Prize 3's precise formulation. Wolfram's Prize 3 asks specifically whether a finite machine receiving index n as input can compute the *fixed* single-black-cell center value $c[n]$ in sub-linear time. Our essentiality result is stated for an algorithm that takes a general initial configuration c as input, giving an $\Omega(n)$ *query-complexity* lower bound for that model. These are different computational problems.

Bridging to Prize 3 requires one of two independent results, neither of which is established here:

- *Incompressibility of the center column*: prove that the Kolmogorov complexity $K(c[0], \ldots, c[n]) \geq \alpha n$ for some $\alpha > 0$ and all large n. A sub-linear TM would then constitute a compression below this threshold, a contradiction. This is equivalent to Prize 3 restated in information-theoretic language.

- *Circuit/TM lower bound*: establish that no $o(n)$-depth circuit family (or $o(n)$-time Turing machine receiving n as input) computes the center-column value $c[n]$. The $\Omega(n)$ query-complexity lower bound proved here is a clean, *unconditional* result: it does not interact with the Razborov-Rudich natural-proofs barrier [8], which concerns super-polynomial *circuit-size* lower bounds rather than decision-tree depth. The gap is that our computational model (an algorithm reading cells of an explicit initial configuration) differs from Prize 3's model (computing the fixed single-black-cell sequence from index n). Bridging that gap would require techniques outside the sensitivity/essentiality framework, and the RR barrier may then become relevant.

A third candidate path exploits the *3-cell block identity.* Let $e_n = [0, \ldots, 0, 1, 0, \ldots, 0]$ (single 1 at position n, length $2n+1$) be the single-black-cell initial condition that Prize 3 is about, and let $t_n = [0, \ldots, 0, 1, 1, 1, 0, \ldots, 0]$ (three consecutive 1s at positions $n-2$, $n-1$, n, length $2n-1$). One step of Rule 30 from e_n produces exactly t_n: positions touching the single 1 evaluate to $\texttt{rule30}(0,0,1) = \texttt{rule30}(0,1,0) = \texttt{rule30}(1,0,0) = 1$ and all others give 0. Therefore

$$s(n) \;=\; \texttt{rule30}_n(e_n) \;=\; \texttt{rule30}_{n-1}(t_n),$$

where $s(n)$ is the single-black-cell center-column sequence. The causal cone of t_n spans positions 0 through $2(n-1)$ at step 0. However, formalizing a lower bound from this identity faces a concrete obstacle: direct computation shows that *position n of e_n is not always sensitive.* Flipping $e_n[n]$ yields the all-zeros configuration, which always evolves to 0. Therefore $k = n$ is sensitive for e_n if and only if $s(n) = 1$. For the half of n-values where $s(n) = 0$ (e.g. $n \in \{2, 6, 7, 10, 11, 12, 14, 17, 18, 20\}$ in the range 1–20), position n contributes nothing to the cone argument for e_n specifically. A cone lower bound that relies on position n being a hard input for position $k = n$ would not apply to these values of n. Structurally, the cone from t_n is geometrically saturated (information travels at the maximum CA speed of one cell per step), but this is a necessary, not sufficient, condition for lower bounds.

The sequence $s(n)$ does exhibit strong pseudorandomness: Berlekamp-Massey finds no linear recurrence of order ≤ 6; linear complexity is $\approx n/2$ (maximal for an n-bit sequence); and the algebraic normal form of $\texttt{rule30}_n$ has degree $2n-1$ with a unique highest-degree monomial (the product of all interior variables), confirmed computationally for $n \leq 10$. None of these properties individually closes the bridge to Prize 3, but they make it implausible that any short-period recurrence or low-complexity formula computes $s(n)$. The status of Prize 3 is genuinely open.

Our essentiality theorem characterizes the causal structure of Rule 30 precisely and constitutes necessary groundwork; it is not itself sufficient for Prize 3.

Acknowledgments

The author thanks the Lean 4 and Mathlib communities for the verification infrastructure.

References

[1] S. Wolfram, *Random sequence generation by cellular automata*, Advances in Applied Mathematics **7** (1986), 123–169.

[2] G. A. Hedlund, *Endomorphisms and automorphisms of the shift dynamical system*, Mathematical Systems Theory **3** (1969), 320–375.

[3] E. Jen, *Invariant strings and pattern-recognizing properties of one-dimensional cellular automata*, Journal of Statistical Physics **43** (1986), 243–265.

[4] S. Wolfram, *A New Kind of Science*, Wolfram Media, 2002.

[5] E. Rowland, *Local nested structure in rule 30*, Complex Systems **16** (2006), 239–258.

[6] G. Cattaneo, E. Formenti, G. Manzini, L. Margara, *Ergodicity, transitivity, and regularity for linear cellular automata over $\mathbb{Z}_m$*, Theoretical Computer Science **217** (1999), 99–122.

[7] H. Buhrman, R. de Wolf, *Complexity measures and decision tree complexity: a survey*, Theoretical Computer Science **288** (2002), 21–43.

[8] A. A. Razborov, S. Rudich, *Natural proofs*, Journal of Computer and System Sciences **55** (1997), 24–35.

[9] K. Sutner, *On the computational complexity of finite cellular automata*, Journal of Computer and System Sciences **50** (1995), 87–97.

[10] N. Nisan, *CREW PRAMs and decision trees*, SIAM Journal on Computing **20** (1991), 999–1007.

A Lean 4 Source Files

The Lean source is in the repository at `P2p/`:

- `Prize3_Complete.lean` — Core definitions, CA evolution, boundary theorems
- `LiftingLemma_LeftPermutive.lean` — Left-permutivity algebra, `lifting_lemma` axiom, and `rule30_prize3`

To verify: install Lean 4 and Mathlib, then run `lake build`.

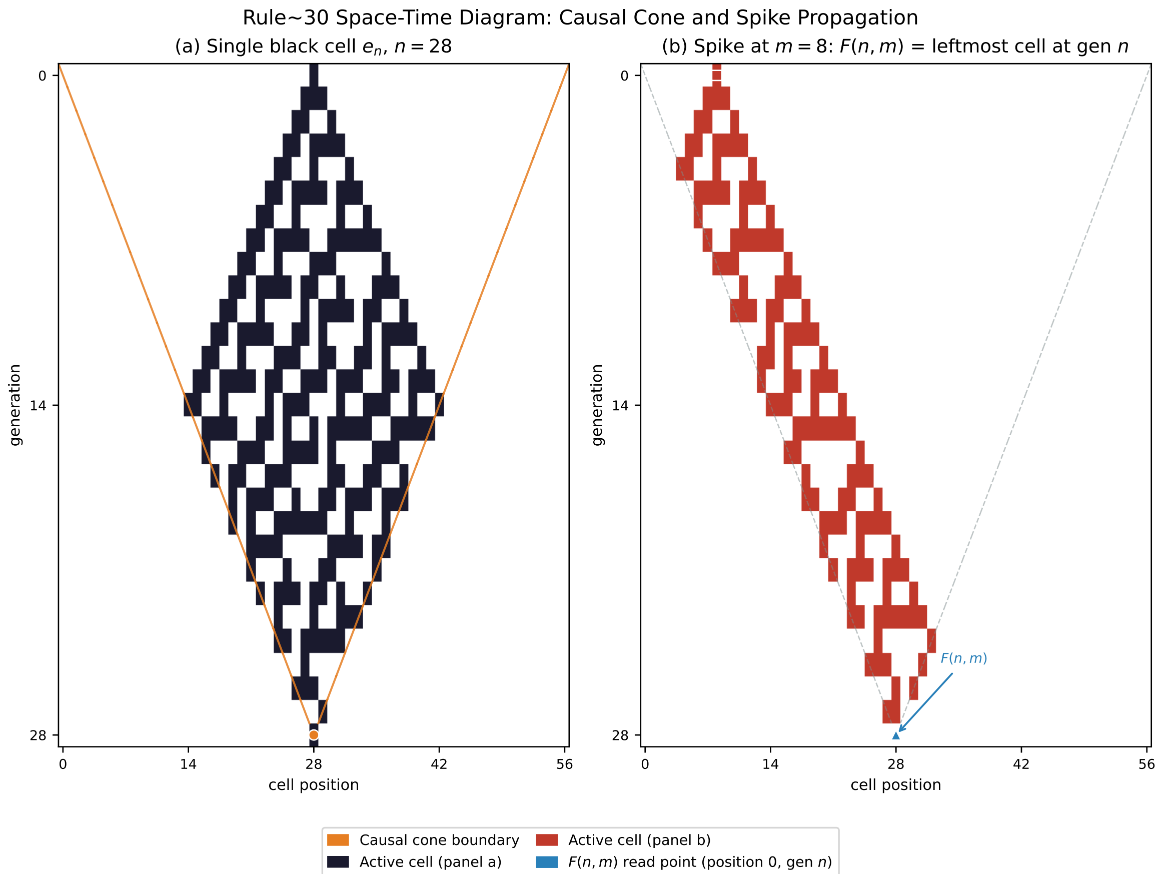

Figure 1: **(a)** Rule 30 space-time diagram from the single-black-cell initial condition e_n ($n = 28$ steps). The orange triangle marks the causal cone boundary, contracting at one cell per step on each side; the orange circle marks the center-cell output $s(n) = \texttt{rule30}_n(e_n)$. Every cell in the cone contributes to this output for some initial configuration (Theorem 1.2). **(b)** Evolution from a spike at fixed position $m = 8$. The blue marker indicates the $F(n, m)$ read point: the leftmost cell of the cone after n steps. The SubcaseB condition $F(n, m) = 0$, $G(n, m) = 1$ is the key open case in the lifting lemma proof; its periodic structure (period $P_m = 2^6$ for $m = 8$) drives the proof plan of Section 4.